Lecture 5:
Finding Features

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Outline

- Local invariant features
  - Motivation
  - Requirements, invariances

- Keypoint Localization
  - Harris corner detector

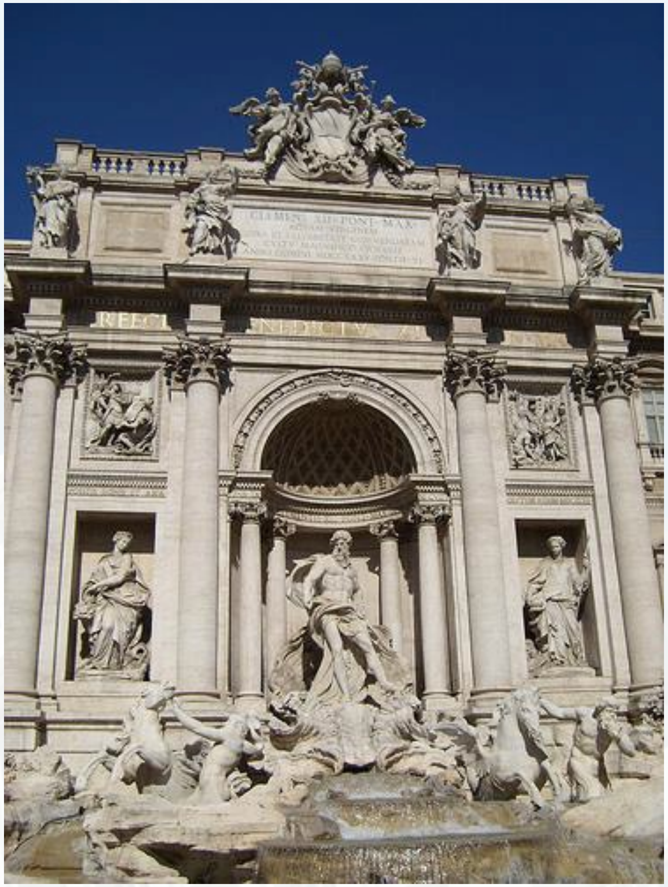
- Scale Invariant Keypoint Detection

Image matching: a challenging problem

Template

Query image

Image matching: a challenging problem

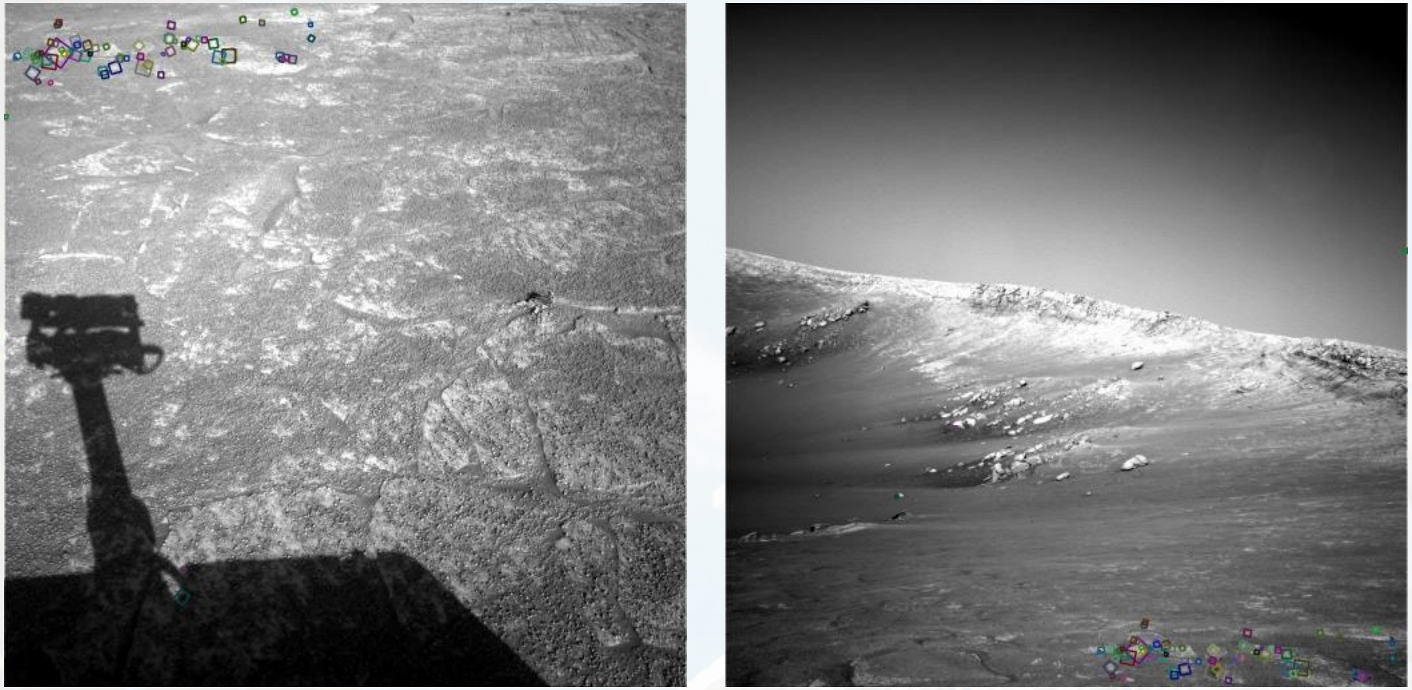


Harder Case

Harder Still

NASA Mars Rover Image

Answer Below (Look for tiny colored squares)

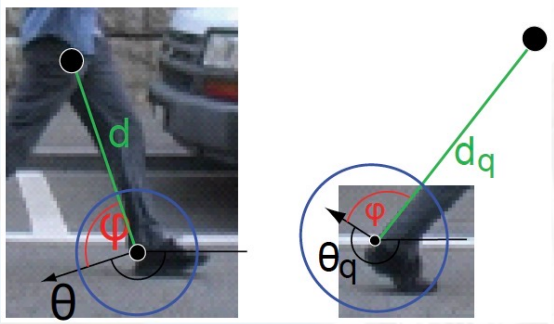


NASA Mars Rover images with SIFT feature matches

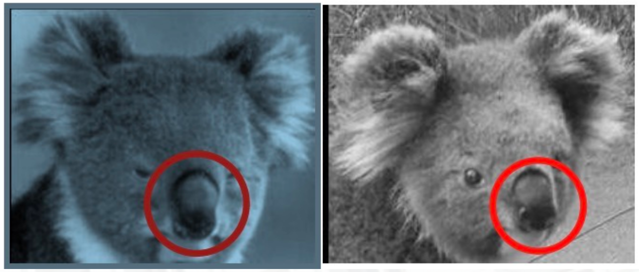
Motivation for using local features

- Global representations have major limitations. Instead, describe and match only local regions.
- Increased robustness to
  - –Occlusions

  - –Articulation

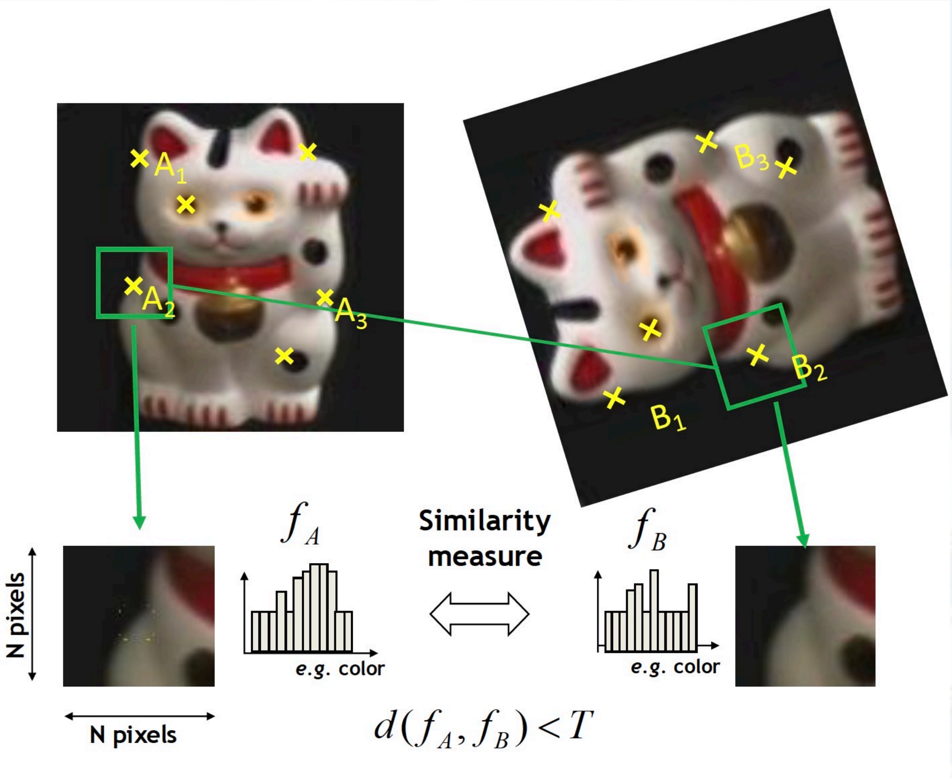


  - –Intra-category variations



General Approach

1. Find a set of distinctive key-points
2. Define a region around each keypoint
3. Extract and normalize the region content
4. Compute a local descriptor from the normalized region
5. Match local descriptors



Common Requirements

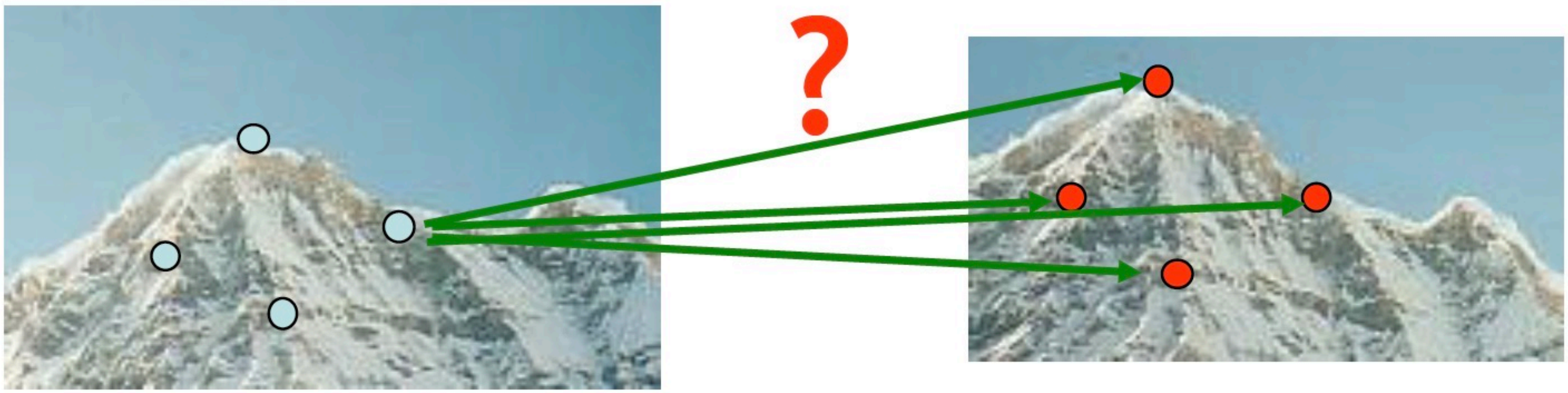
- Problem 1:
  - –Detect the same point independently in both images

No chance to match!

We need a repeatable detector!

Common Requirements

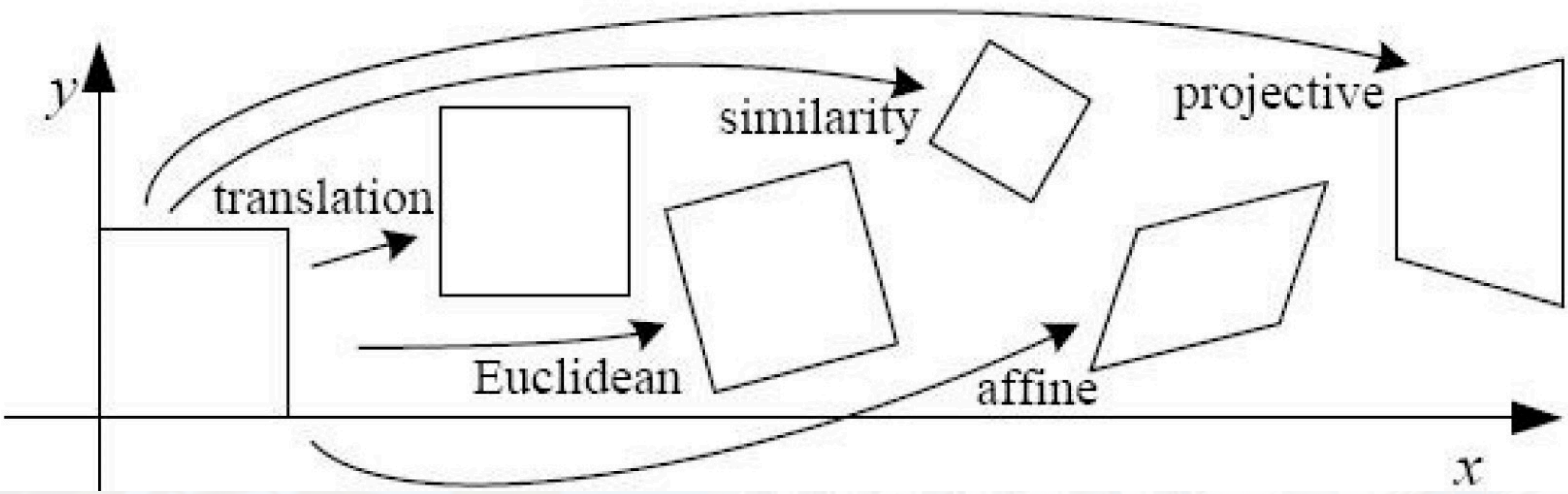
- Problem 1:
  - –Detect the same point independently in both images
- Problem 2:
  - –For each point correctly recognize the corresponding one



We need a reliable and distinctive descriptor!

Feature Invariances: Geometric Transformations

Levels of Geometric Invariance



Feature Invariances: Photometric Transformations

Requirements for Local Features

- Region extraction needs to be repeatable and accurate
  - Invariant to translation, rotation, scale changes
  - Robust or covariant to out-of-plane (affine) transformations
  - Robust to lighting variations, noise, blur, quantization

- Locality: Features are local, therefore robust to occlusion and clutter.

- Quantity: We need a sufficient number of regions to cover the object.

- Distinctiveness: The regions should contain “interesting” structure.

- Efficiency: Close to real-time performance.

Many Existing Feature Detectors Available

•Hessian & Harris	[Beaudet ‘78], [Harris ‘88]
•Laplacian, DoG	[Lindeberg‘98], [Lowe ‘99]
•Harris-/Hessian-Laplace	[Mikolajczyk& Schmid ‘01]
•Harris-/Hessian-Affine	[Mikolajczyk& Schmid ‘04]
•EBR and IBR	[Tuytelaars& Van Gool ‘04]
•MSER	[Matas‘02]
•Salient Regions	[Kadir & Brady ‘01]
•Others…	

Those detectors have become a basic building block for many applications in Computer Vision.

Outline

- Local invariant features
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  – Requirements, invariances

- Keypoint Localization
  – Harris corner detector

- Scale Invariant Keypoint Detection

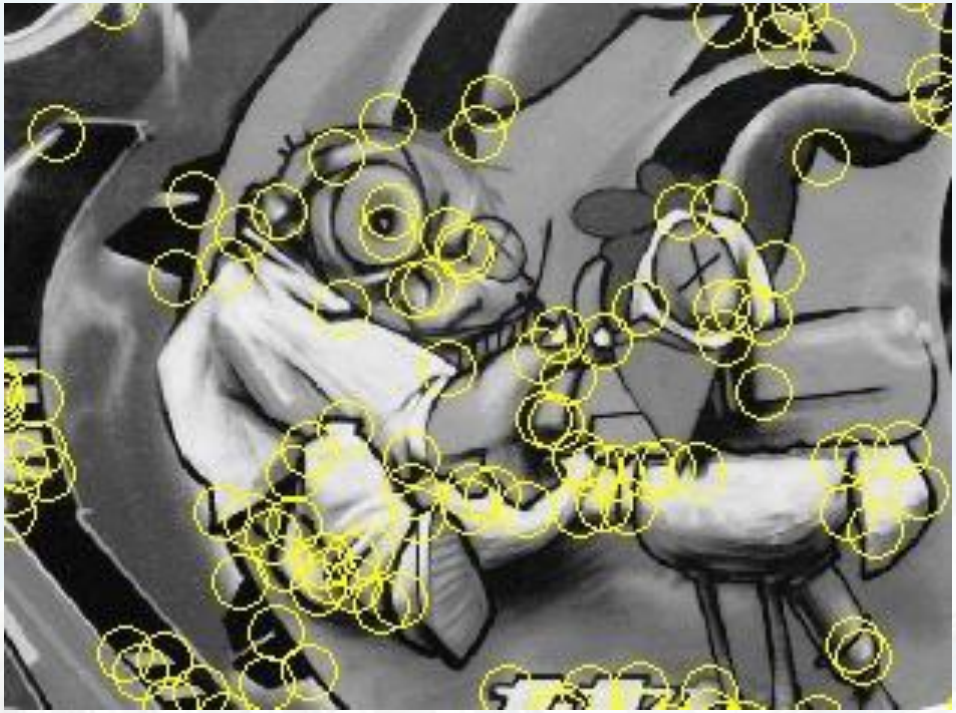
Keypoint Localization

Goals:

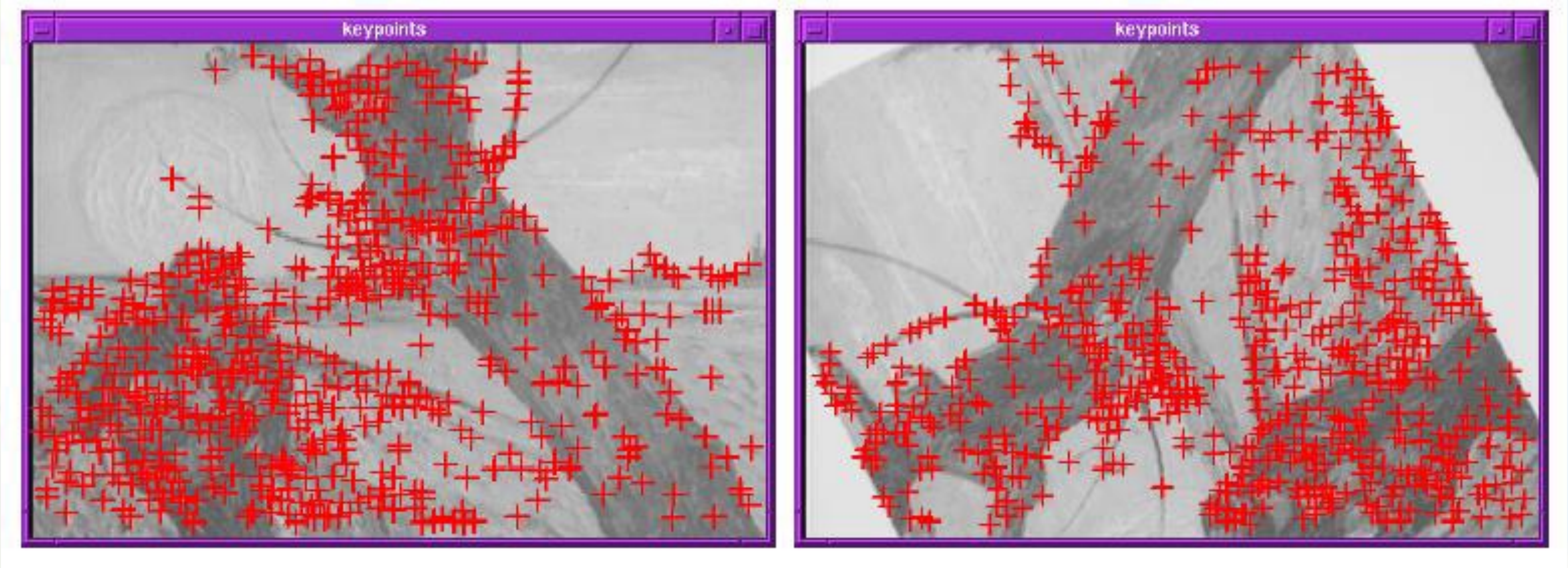
- –Repeatable detection
- –Precise localization
- –Interesting content

Solution:

- – Look for two-dimensional signal changes



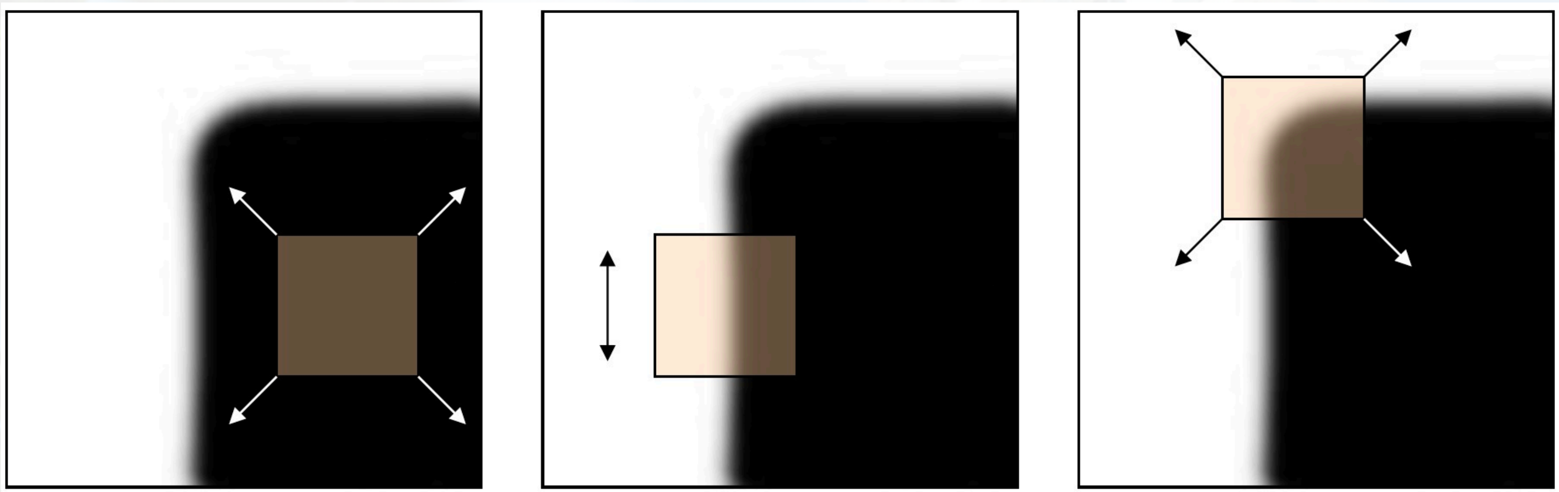
Finding Corners



- Key property:
  - In the region around a corner, the image gradient has two or more dominant directions
- Corners are ***repeatable*** and ***distinctive***

Corners as Distinctive Interest Points

- Design criteria
  - –We should easily recognize the corner point by looking through a small window (locality)
  - –Shifting the window in any direction should give a large change in intensity (good localization)



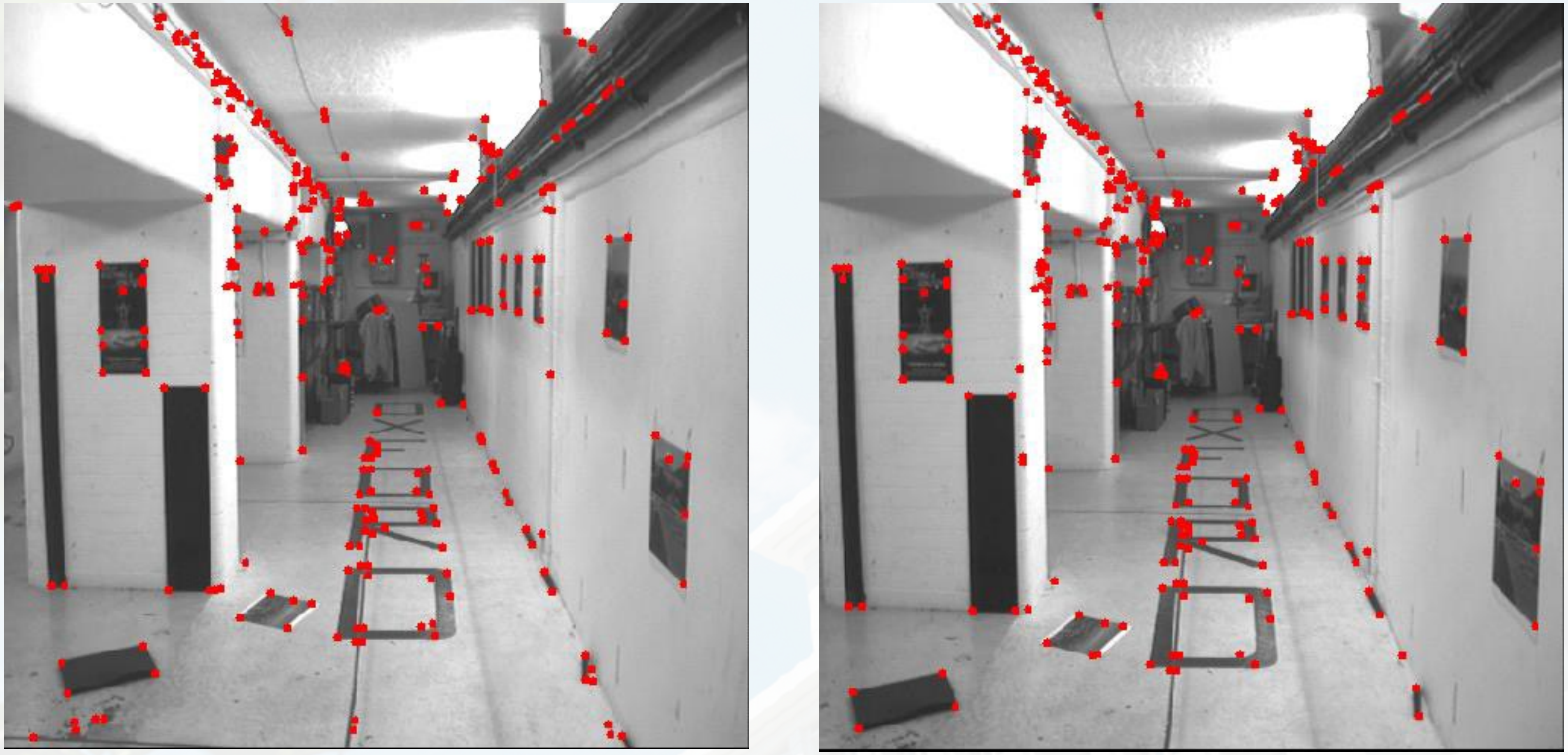
"flat" region:
no change in all directions

"edge":
no change along the edge direction

"corner":
significant change in all directions

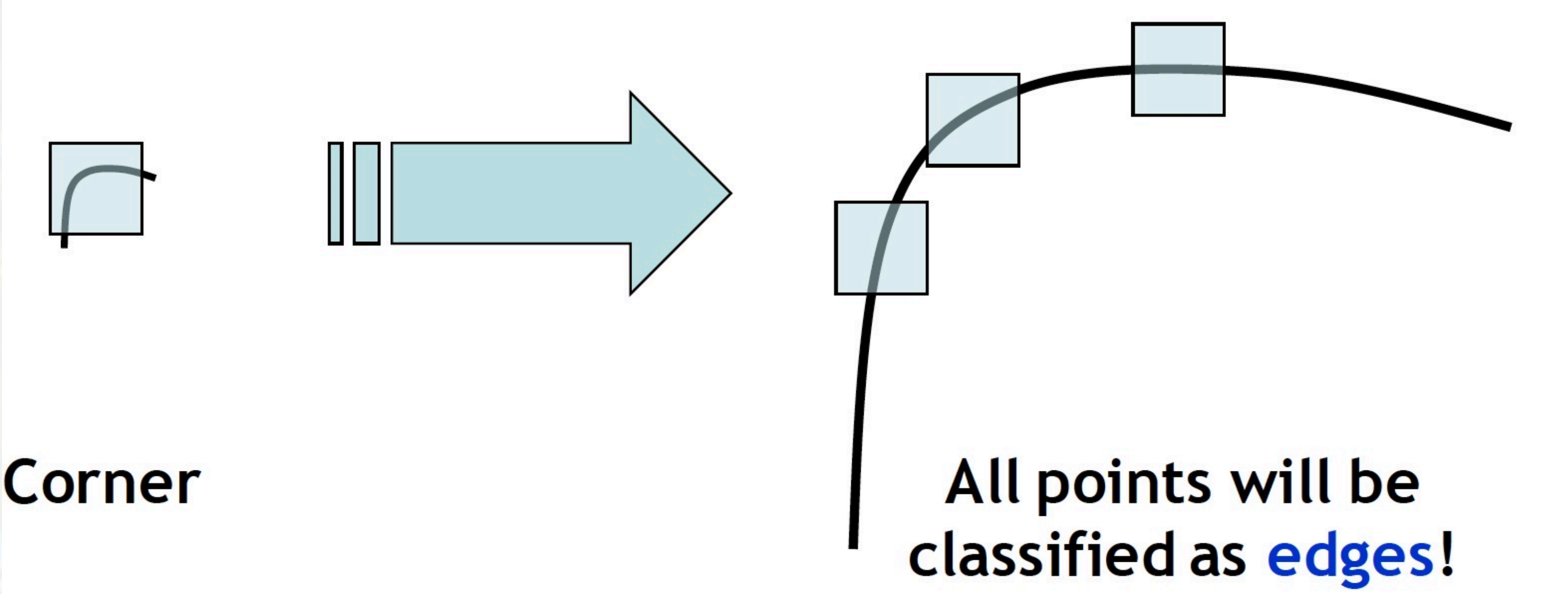
Harris Detector: Example

Harris Detector: Example



•Results are well suited for finding stereo correspondences

Harris Detector
– Not Invariant to Image Scale



Not invariant to image scale!

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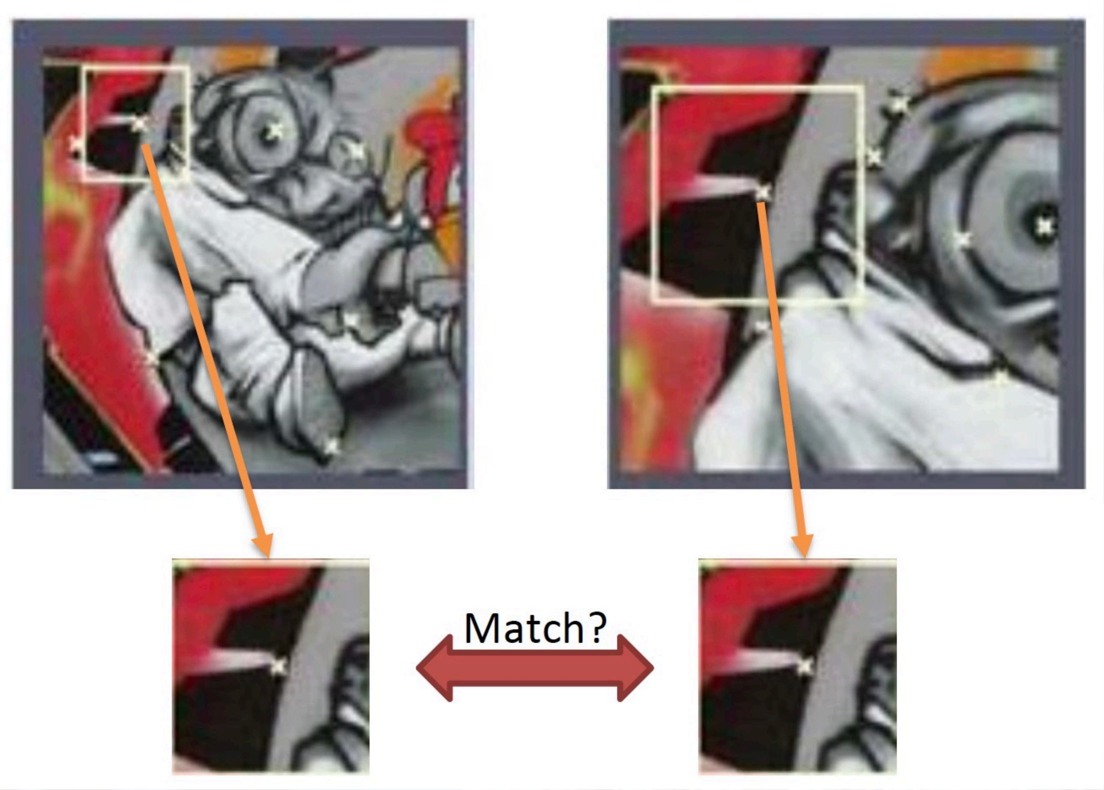
- Scale Invariant Keypoint Detection

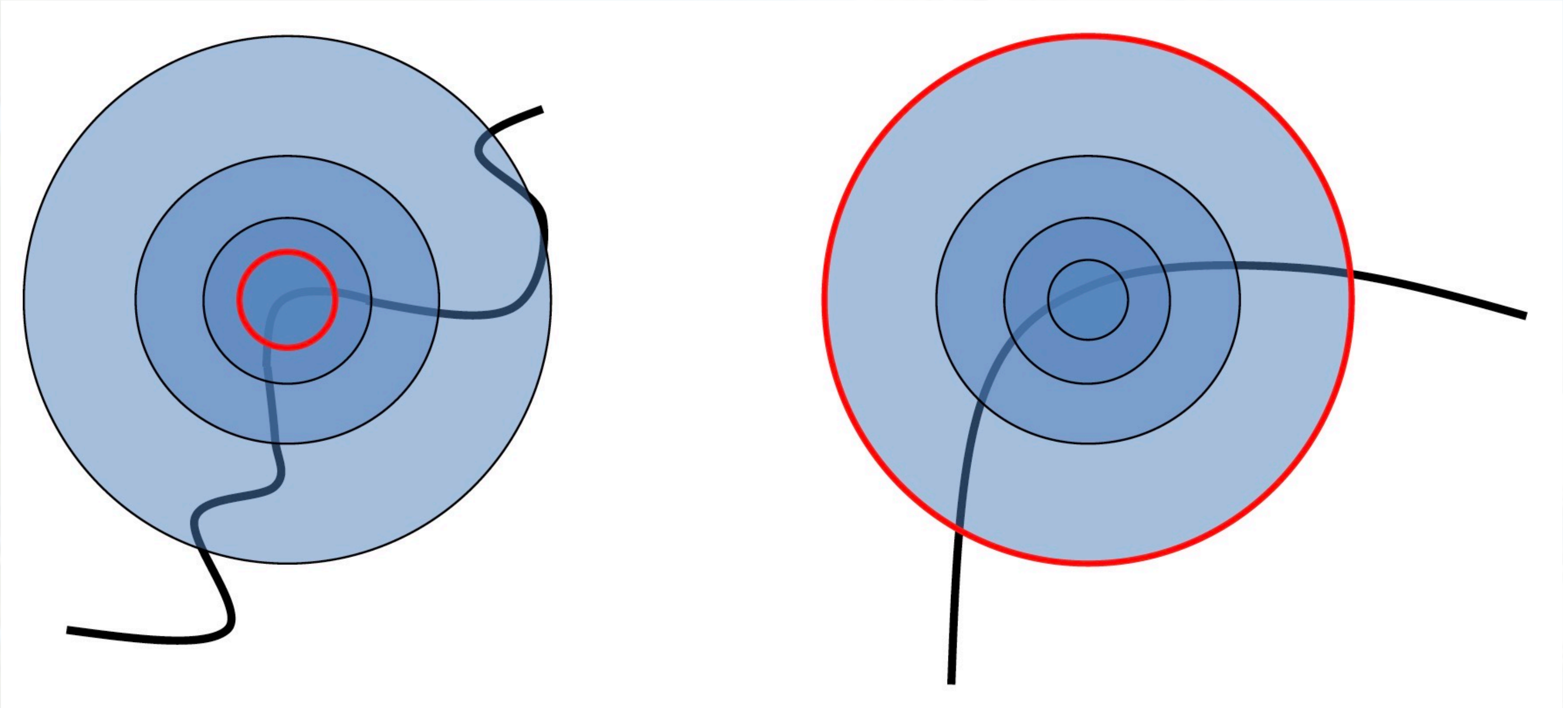
Does Scale Matter?

• For feature matching, it is important to estimate the size of the neighborhood that can lead to best matching between images of different scale.



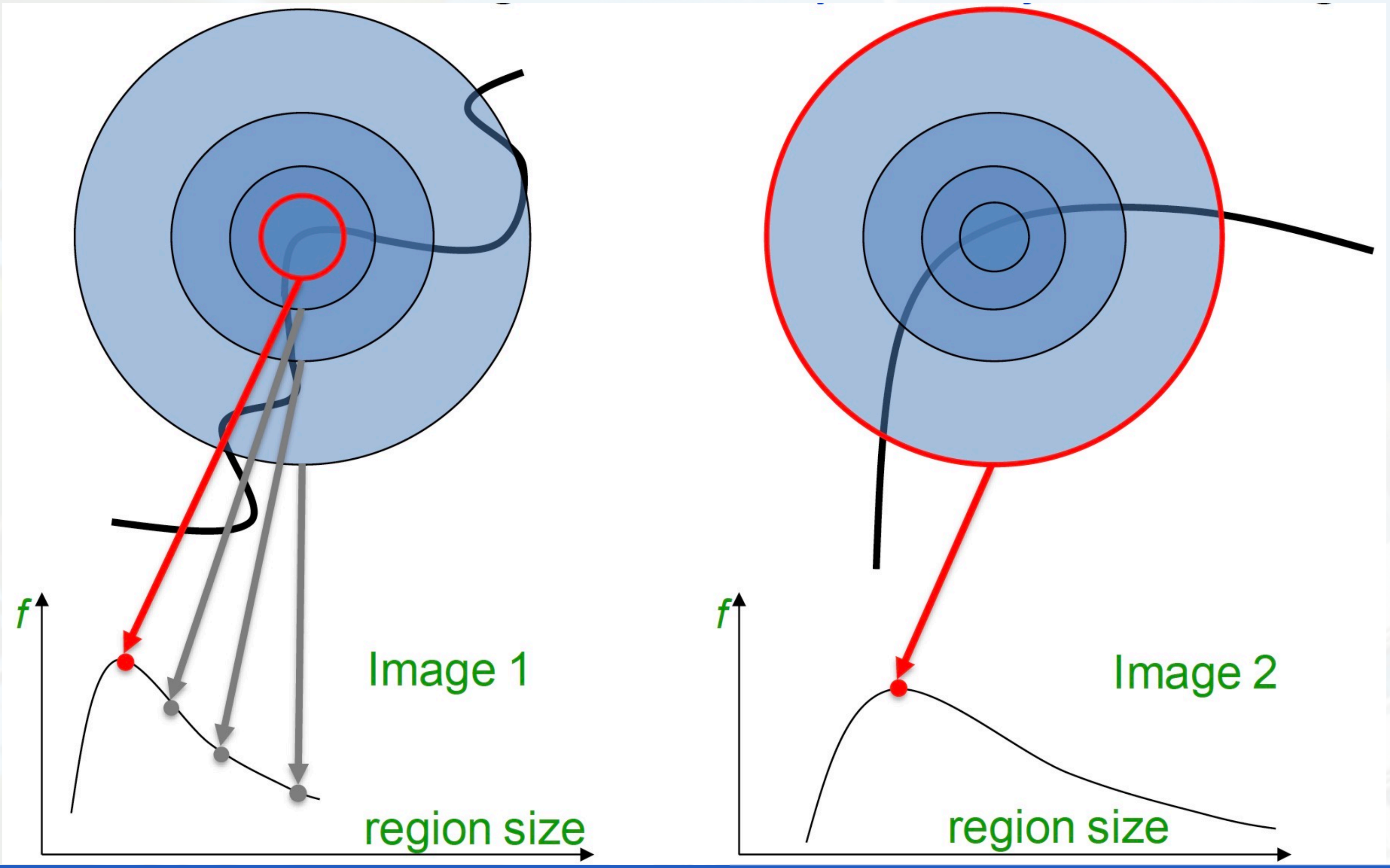
Scale Invariant Detection

- Consider regions (e.g. circles) of different sizes around a point
- What region size do we choose, so that the regions look the same in both images?



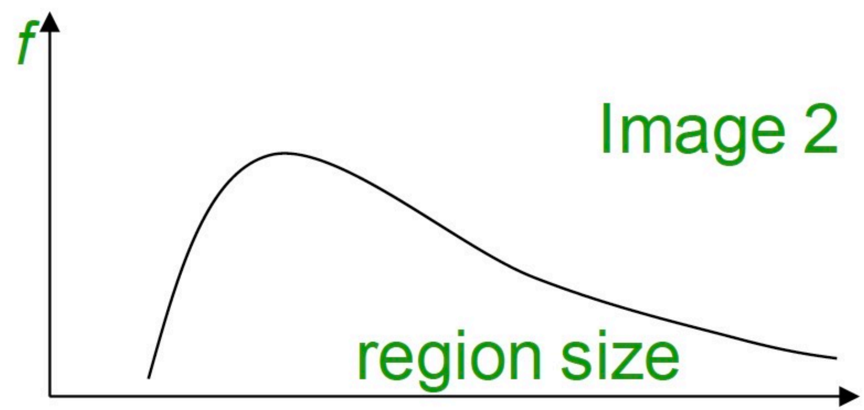
Scale Invariant Detection

- Problem: How do we choose region sizes ***independently*** in each image?



Scale Invariant Detection

- Solution:
  - Design a function on the region (circle), which is "scale invariant": has the same value for corresponding regions, even if they are at different scales
    - Example: average intensity. For corresponding regions (even of different sizes) it will be the same.
  - Given a point in one image, we can think of it as a function of region size (circle radius)

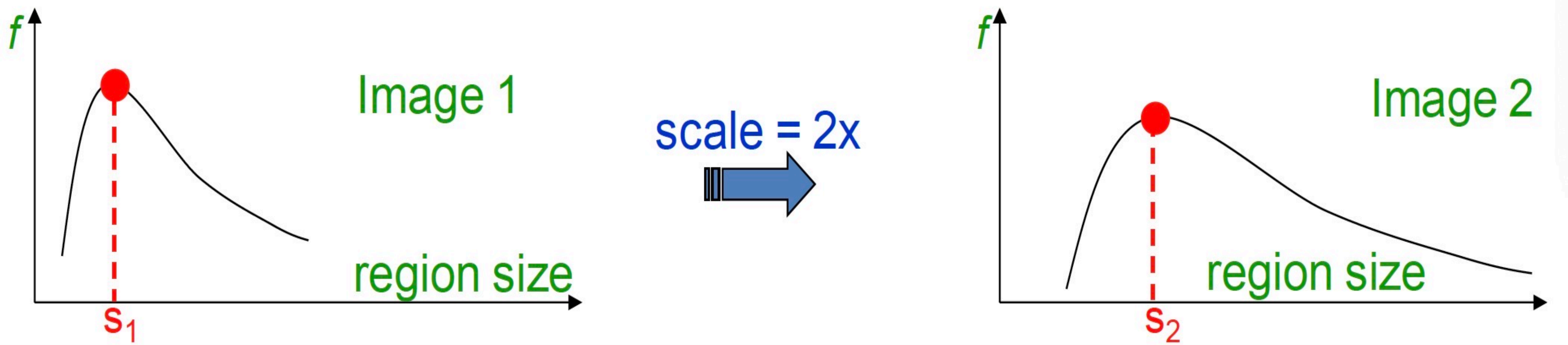


Scale Invariant Detection

- Common approach:
  - Take a local maximum of this function

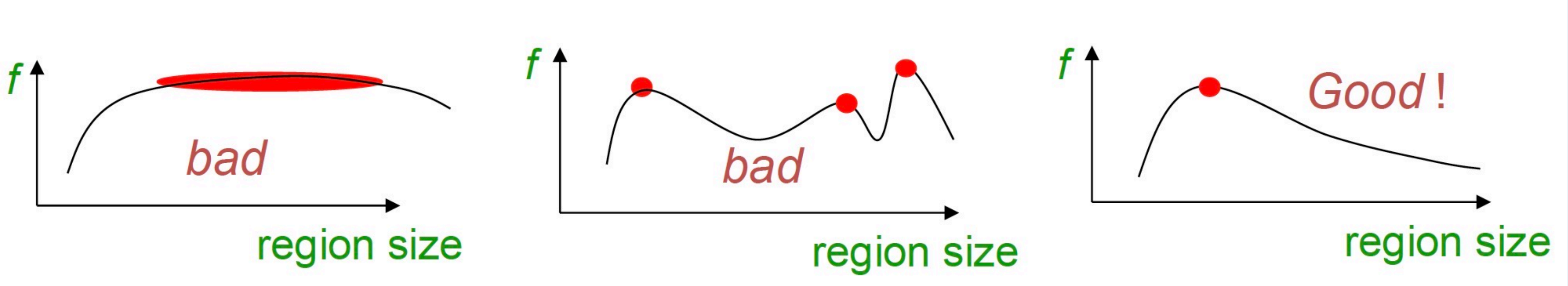
- Important: this scale invariant region size is found in each image independently!

- Observation: region size, for which the maximum is achieved, should be co-variant with image scale.



Scale Invariant Detection

- A "good" function for scale selection has one stable sharp peak



- For usual images: a good function would be one which responds to contrast (sharp local intensity change)

Scale Invariant Detection

- Functions for determining scale: f = Kernel * Image

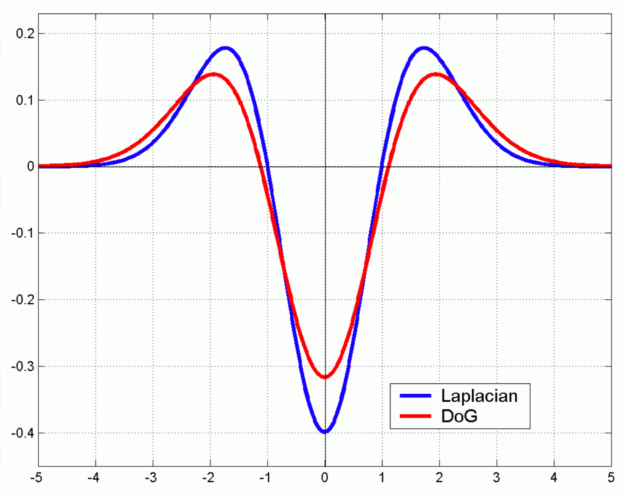
- Choices of Kernels

  – Laplacian:

$$L = \sigma^2 \left(G_{xx}(x,y,\sigma) + G_{yy}(x,y,\sigma) \right)$$

  – Difference of Gaussians (DoG):

$$DoG = G(x,y,k\sigma) - G(x,y,\sigma)$$



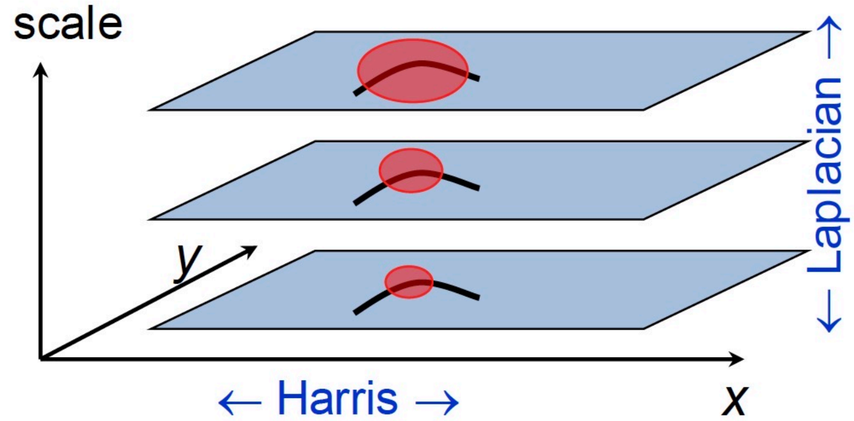
Where G is the Gaussian function

$$G(x,y,\sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{x^2+y^2}{2\sigma^2}}$$

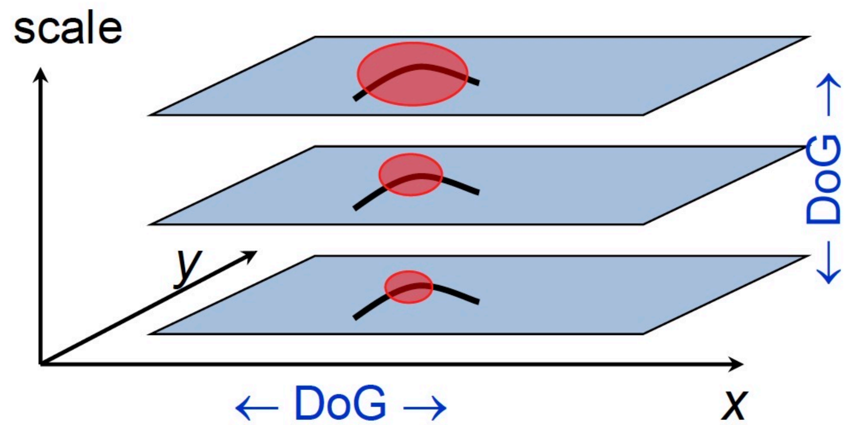
Note: both kernels are invariant to *scale* and *rotation*

Scale Invariant Detection

- Harris-Laplacian[1]
  Find local maximum of:
  - Harris corner detector in space (image coordinates)
  - Laplacian in scale



- DoG (from SIFT by Lowe)[2]
  Find local maximum of:
  - Difference of Gaussians in space and scale



[1] K.Mikolajczyk, C.Schmid. "Indexing Based on Scale Invariant Interest Points". ICCV 2001
[2] D.Lowe. "Distinctive Image Features from Scale-Invariant Keypoints". IJCV 2004

Content

- Introduction of SIFT
  - **Scale Space**
  - **Scale space peak selection**
  - **Key point localization**
  - **Orientation Assignment**
  - **Key point descriptor**
  - **Key point matching**
  - **HOG**
  - **LBP**

Introduction of SIFT

David Lowe

Introduction of SIFT

- Stands for Scale Invariant Feature Transform
- Patented by university of British Columbia
- Similar to the one used in primate visual system (human, ape, monkey, etc.)
- Transforms image data into scale-invariant coordinates

D. Lowe. *Distinctive image features from scale-invariant key points.*, International Journal of Computer Vision 2004.

Introduction of SIFT

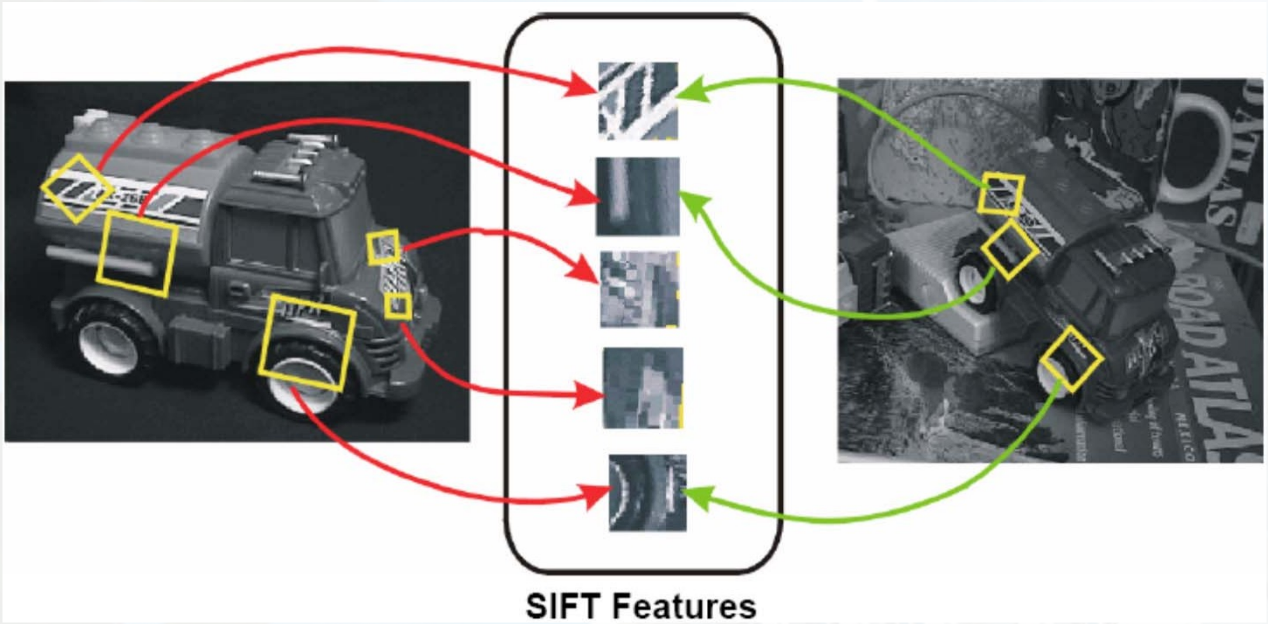
- Extracting distinctive invariant features
  - Correctly matched against a large database of features from many images
- Invariance to image scale and rotation
- Robustness to
  - Affine distortion
  - Change in 3D viewpoint
  - Addition of noise
  - Change in illumination

Introduction of SIFT

Advantages：

- **Locality:** features are local, so robust to occlusion and clutter
- **Distinctiveness:** individual features can be matched to a large database of objects
- **Quantity:** many features can be generated for even small objects
- **Efficiency:** Not suitable for real-time tasks, such as SLAM

Introduction of SIFT



Invariant Local Features

Introduction of SIFT

Steps of SIFT：

- Scale space peak selection
  - Potential locations for finding features
- Key point localization
  - Accurately locating the feature key points
- Orientation assignment
  - Assigning orientation to the key points
- Key point descriptor
  - Describing the key point as a high dimensional vector

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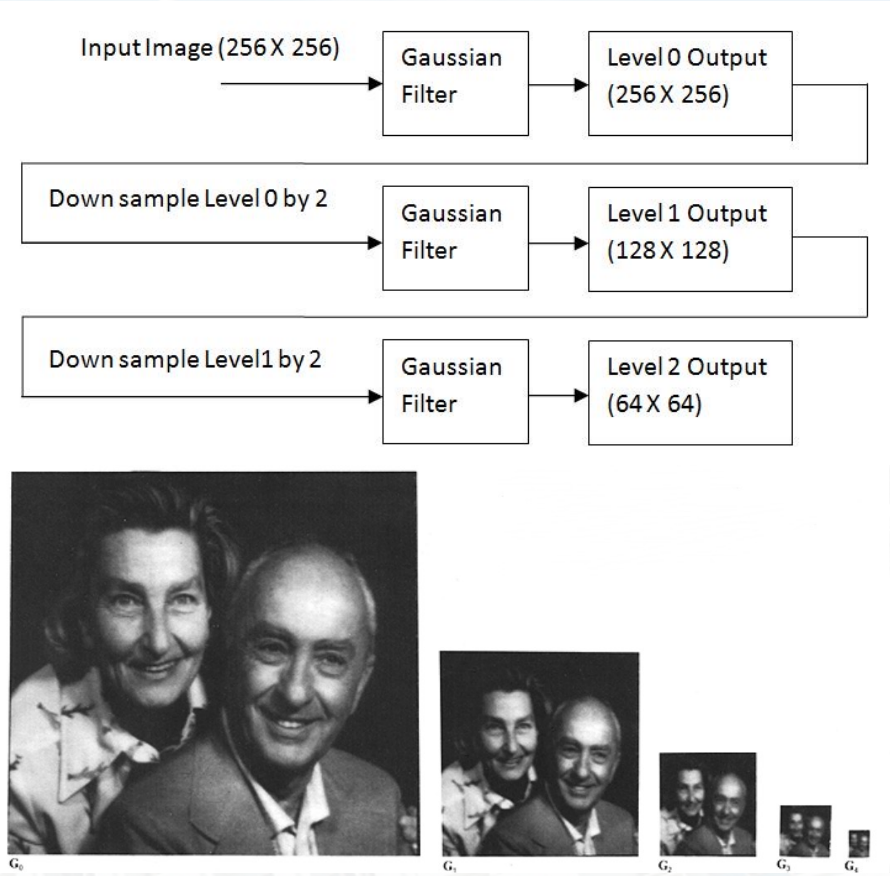
Scale Space

Gaussian Pyramid :

1. Take input image, apply Gaussian filter, output will be level 0 image.
2. Down-sample the level 0 image by 2, apply Gaussian filter, output will be level 1 image.
3. Down-sample the level 1 image by 2, apply Gaussian filter, output will be level 2 image.
4. Similar way we can create full pyramid of Gaussian.

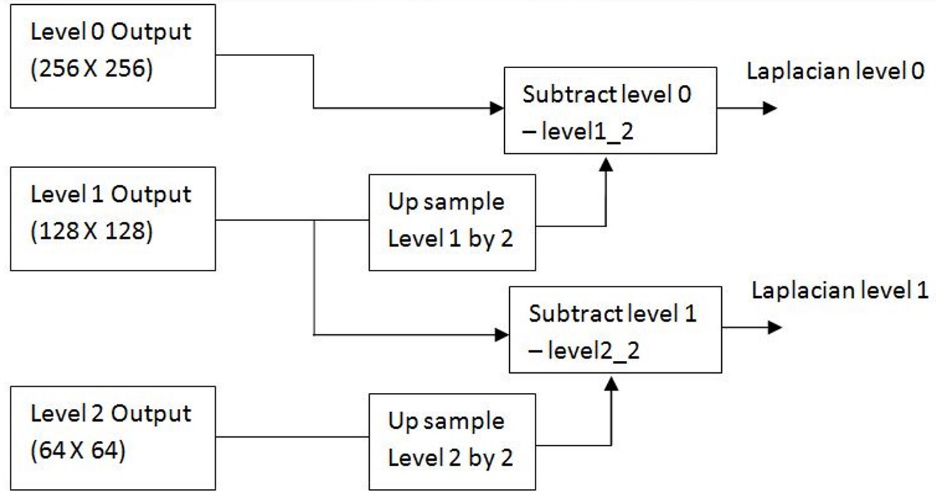


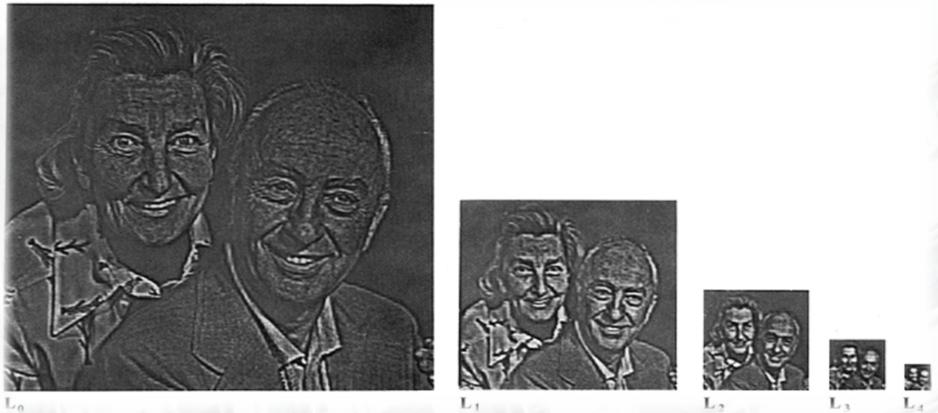
Scale Space

Laplacian Pyramid：

Suppose Level 0, level 1 and level 2 is the output of Gaussian pyramid as we discussed above.

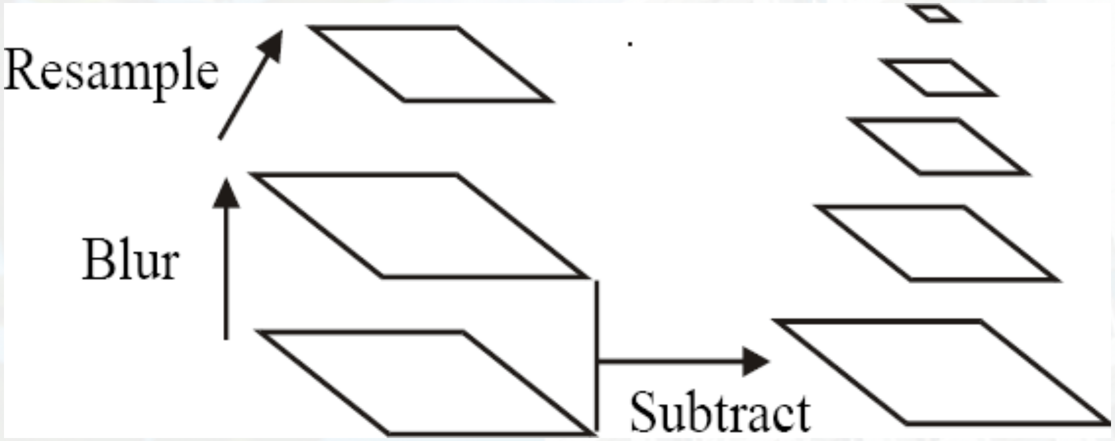
1. Up-sample the Gaussian level 1 image by 2 (level 1_2)and subtract the level 1_2 image from level 0 image, output will be Laplacian of level 0.
2. Up-sample the Gaussian level 2 image by 2 (level 2_2) and subtract the level 2_2 image from level 1 Gaussian image, output will be Laplacian of level 1.





Scale Space

- All scales must be examined to identify scale-invariant features
- An efficient function is to compute the Laplacian Pyramid (Difference of Gaussian) (Burt & Adelson, 1983)



Scale Space

$$\frac{\partial G}{\partial \sigma} = \sigma \Delta^2 G$$

Heat Equation

$$\sigma \Delta^2 G = \frac{\partial G}{\partial \sigma} = \frac{G(x, y, k\sigma) - G(x, y, \sigma)}{k\sigma - \sigma}$$

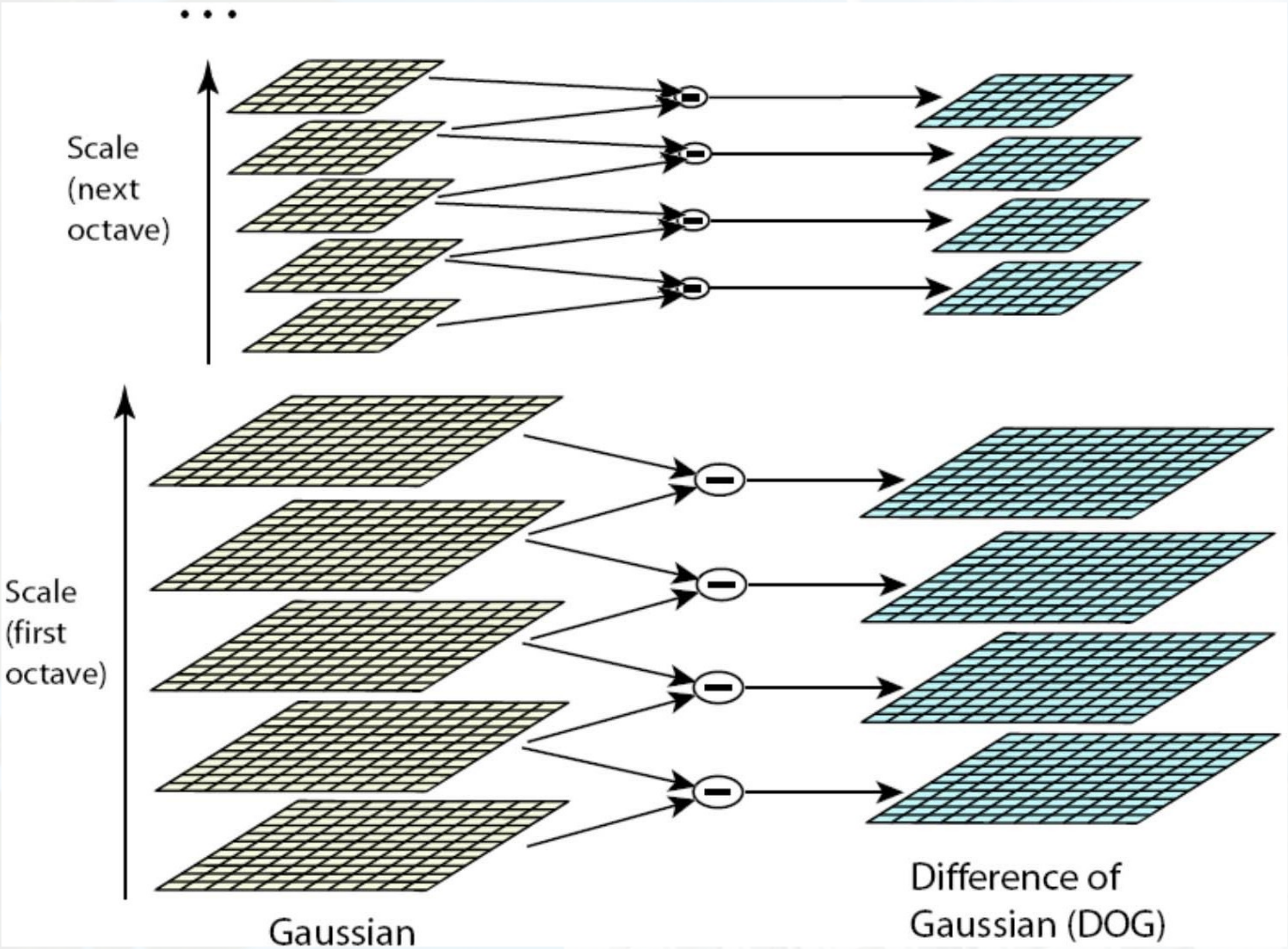
$$G(x, y, k\sigma) - G(x, y, \sigma) \approx (k-1)\sigma^2 \Delta^2 G$$

$$\text{Typical values}: \sigma = 1.6;\ \ k = \sqrt{2}$$

Approximation of LoG by Difference of Gaussians

Scale Space

$$G(x,y,k\sigma) = \frac{1}{2\pi(k\sigma)^2} e^{-(x^2+y^2)/2k^2\sigma^2}$$



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- Scale space peak selection

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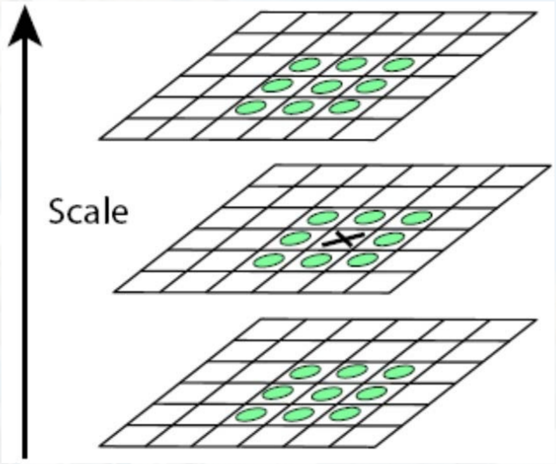
Scale Space Peak Detection

- Compare a pixel (**X**) with 26 pixels in current and adjacent scales (Green Circles)
- Select a pixel (**X**) if larger/smaller than all 26 pixels
- Large number of extrema, computationally expensive



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Key Point Localization

- Candidates are chosen from extrema detection

Key Point Localization

Initial Outlier Rejection:

- Reject Low contrast candidates (sensitive to noise)
- Taylor series expansion of DoG, **D**.

$$D(x) = D + \frac{\partial D^T}{\partial x}\Delta x + \frac{1}{2}\Delta x^T \frac{\partial^2 D}{\partial x^2}\Delta x$$

Δx is the offset

- Deriving the formula and making it 0 $\Delta x = -\frac{\partial^2 D^{-1}}{\partial x^2}\frac{\partial D(x)}{\partial x}$
- Minima or maxima is $D(\hat{x}) = D + \frac{1}{2}\frac{\partial D^T}{\partial x}\hat{x}$
- Value of D*(*x*)* at minima/maxima must be large, $|D(x)|>th$.

Key Point Localization

from 832 key points to 729 key points, th=0.03.

Key Point Localization

Further Outlier Rejection:

- DOG has strong response along edge, so reject poorly localized candidates along an edge
- Assume DOG as a surface
  - Compute principal curvatures (PC)
  - Along the edge one of the PC is very low, across the edge is high

Key Point Localization

Further Outlier Rejection:

- Analogous to Harris corner detector
- Compute Hessian of D

$$\mathbf{H} = \begin{bmatrix} D_{xx} & D_{xy} \\ D_{xy} & D_{yy} \end{bmatrix}$$

$$Tr(H) = D_{xx} + D_{yy} = \lambda_1 + \lambda_2$$

$$Det(H) = D_{xx}D_{yy} - (D_{xy})^2 = \lambda_1 \lambda_2$$

- Remove outliers by evaluating

$$\frac{Tr(H)^2}{Det(H)} = \frac{(r+1)^2}{r} \qquad r = \frac{\lambda_1}{\lambda_2}$$

Key Point Localization

Further Outlier Rejection:

- Following quantity is minimum when r=1
- It increases with r

$$\frac{Tr(H)^2}{Det(H)} = \frac{(r+1)^2}{r}$$

- Eliminate key points if $r > 10$

Key Point Localization

from 729 key points to 536 key points

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Orientation Assignment

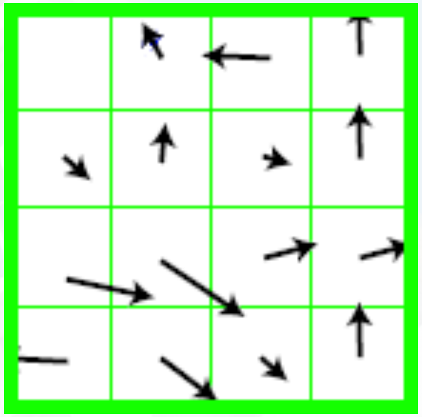
- To achieve rotation invariance
- Compute central derivatives, gradient magnitude and direction of *L* (smooth image) at the scale of key point (x,y)

$$m(x,y) = \sqrt{[L(x+1,y) - L(x-1,y)]^2 + [L(x,y+1) - L(x,y-1)]^2}$$

$$\theta(x,y) = \arctan \frac{L(x,y+1) - L(x,y-1)}{L(x+1,y) - L(x-1,y)}$$

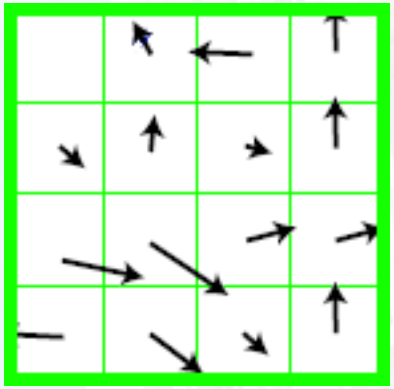
Orientation Assignment

- Create a weighted direction histogram in a neighborhood of a key point (36 bins)
- Weights are
  - Gradient magnitudes
  - Spatial gaussian filter with σ=1.5 x <scale of key point>



Orientation Assignment

- Select the peak as direction of the key point
- Introduce additional key points (same location) at local peaks (within 80% of max peak) of the histogram with different directions



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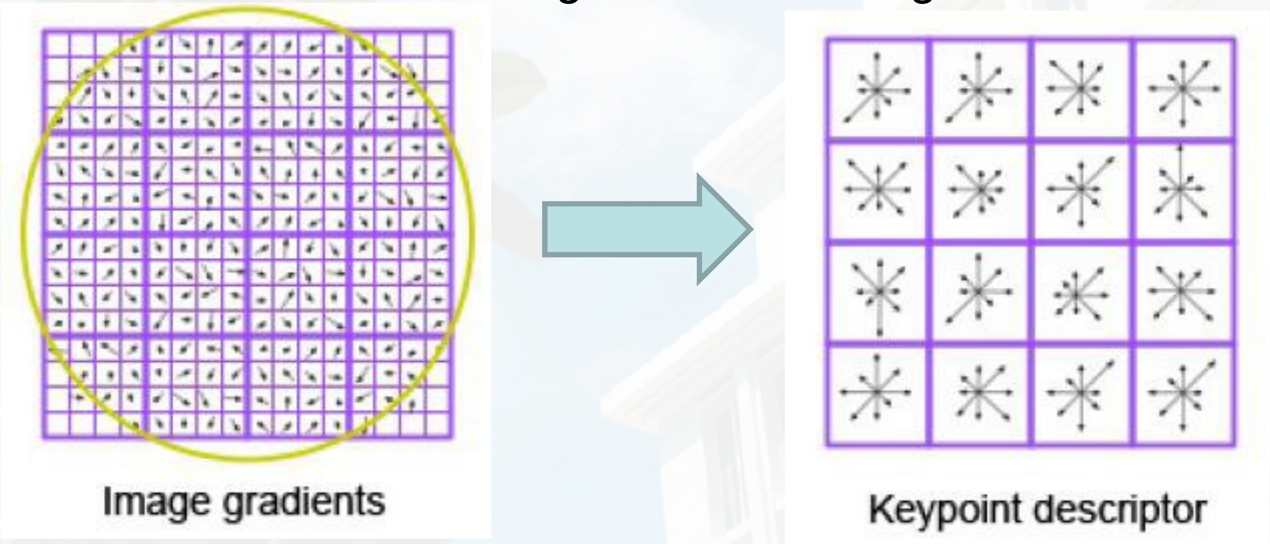
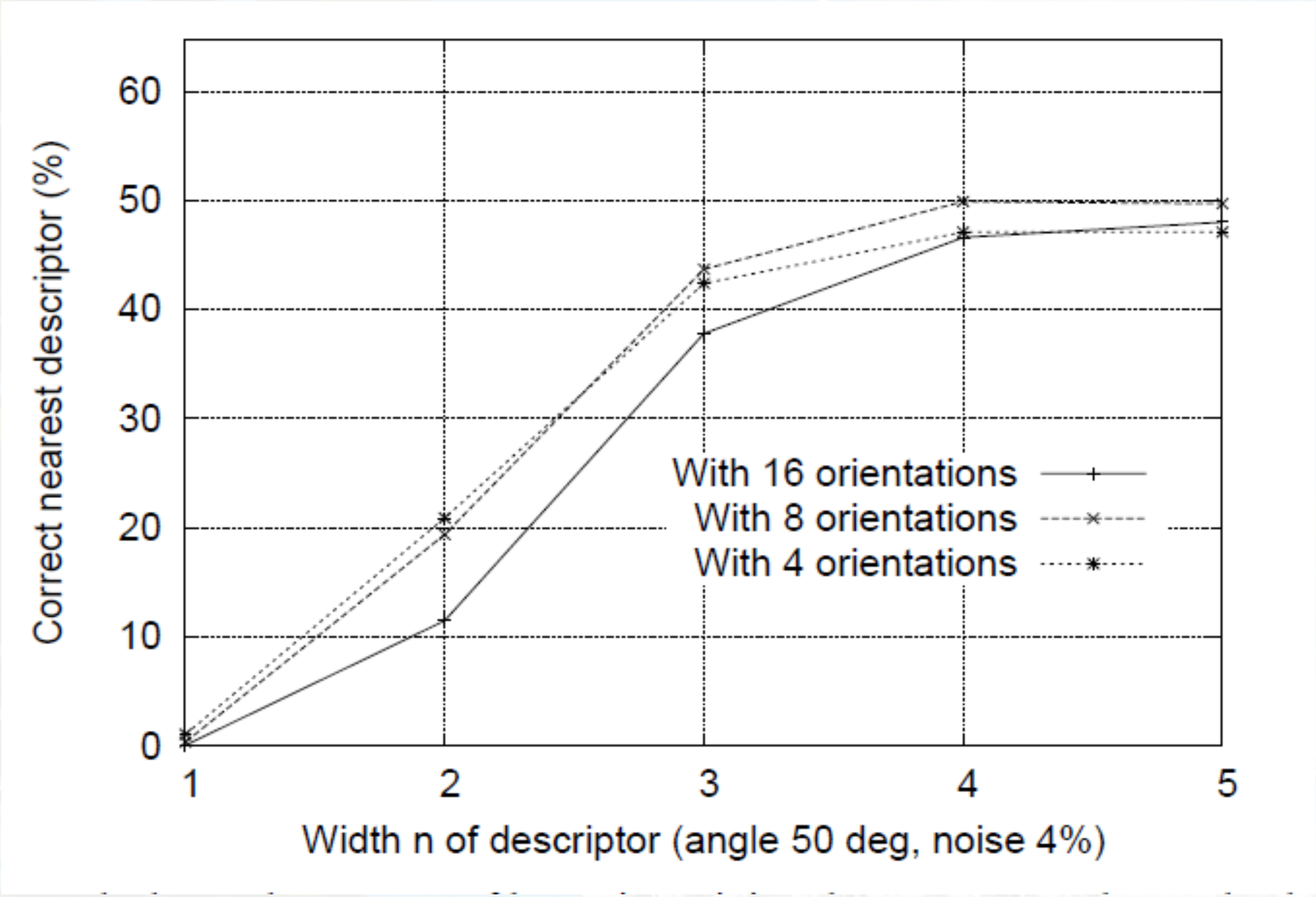
Key point descriptor

- Compute relative orientation and magnitude in a 16x16 neighborhood at key point
- Form weighted histogram (8 bin) for 4x4 regions
  - Weight by magnitude and spatial Gaussian
  - Concatenate 16 histograms in one long vector of 128 dimensions



16 histograms x 8 orientations = 128 features

Key point descriptor



Descriptor Regions (n by n)

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Key point matching

- Match the key points against a database of that obtained from training images.
- Find the nearest neighbor i.e. a key point with minimum Euclidean distance.
  - Efficient Nearest Neighbor matching
    - Looks at ratio of distance between best and 2nd best match (0.8)

Key point matching

Matching local features

Key point matching

Image 1 Image 2

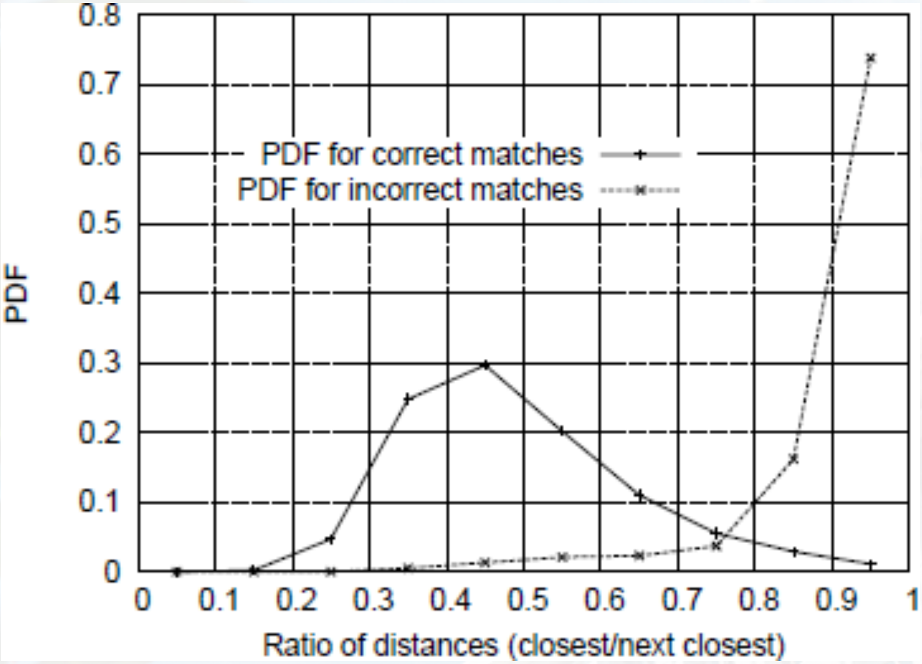
- To generate **candidate matches**, find patches that have the most similar appearance or SIFT descriptor
- Simplest approach: compare them all, take the closest (or closest k, or within a thresholded distance)

Key point matching

Image 1 Image 2

- At what distance do we have a good match?
- To add robustness to matching, can consider **ratio** : distance to best match / distance to second best match
- If low, first match looks good.
- If high, could be ambiguous match.

Key point matching



The ratio of distance from the closest to the distance of the second closest

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HOG

- HOG feature extraction
  - Compute centered horizontal and vertical gradients with no smoothing
  - Compute gradient orientation and magnitudes
    - For color image, pick the color channel with the highest gradient magnitude for each pixel.
  - For a 64x128 image, divide the image into 16x16 blocks of 50% overlap.
    - 7x15=105 blocks in total
  - Each block should consist of 2x2 cells with size 8x8.
  - Quantize the gradient orientation into 9 bins
    - The vote is the gradient magnitude
    - Interpolate votes between neighboring bin center.
    - The vote can also be weighted with Gaussian to down-weight the pixels near the edges of the block.
  - Concatenate histograms (Feature dimension: 105x4x9 = 3,780)

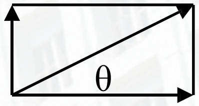
Computing Gradients

- Centered: $$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x-h)}{2h}$$
- Filter masks in x and y directions
  - Centered:

-1	0	1

-1
0
1

- Gradient
  - Magnitude: $s = \sqrt{s_x^2 + s_y^2}$

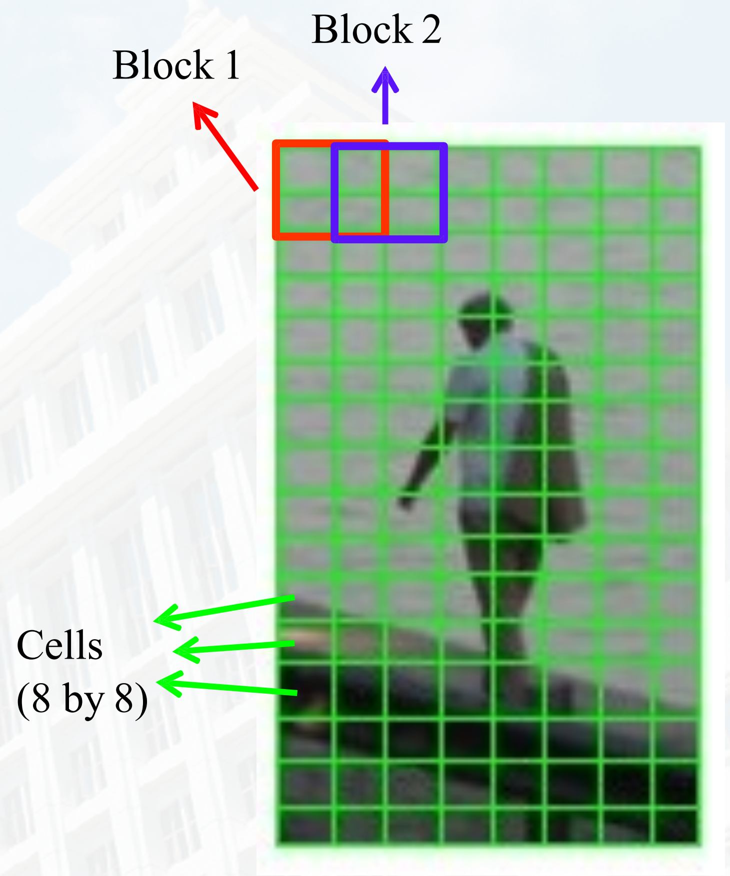


  - Orientation: $\theta = \arctan(\frac{s_y}{s_x})$

HOG

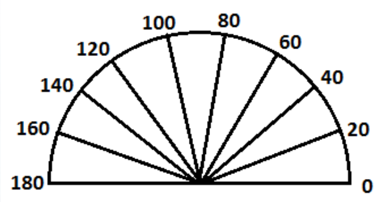
Blocks, Cells

- 16x16 blocks of 50% overlap.
  - 7x15=105 blocks in total
- Each block should consist of 2x2 cells with size 8x8.

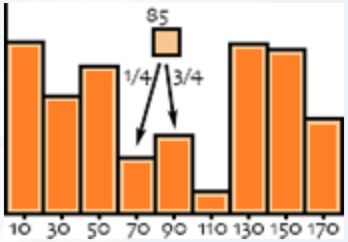


HOG

- Each block consists of 2x2 cells with size 8x8
- Quantize the gradient orientation into 9 bins (0-180)
  - The vote is the gradient magnitude
  - Interpolate votes linearly between neighboring bin centers.
    - Example: if θ=85 degrees.
    - Distance to the bin center Bin 70 and Bin 90 are 15 and 5 degrees, respectively.
    - Hence, ratios are 5/20=1/4, 15/20=3/4.



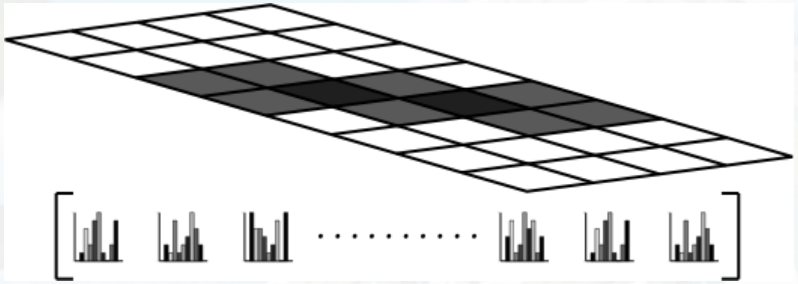
9 Bins



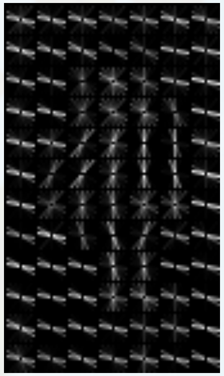
Bin centers

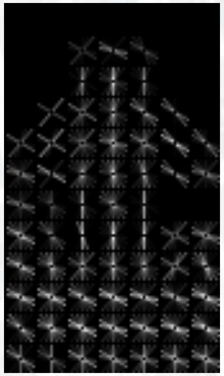
Final Feature Vector

- Concatenate histograms
  - Make it a 1D vector of length 3780.



- Visualization





HOG

Reference

Navneet Dalal and Bill Triggs “Histograms of Oriented Gradients for Human Detection” CVPR05

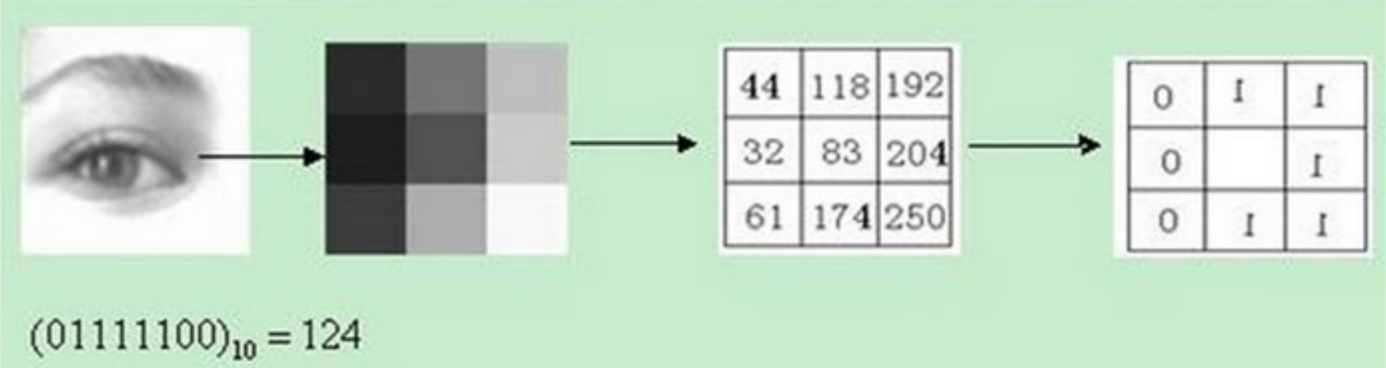
Content

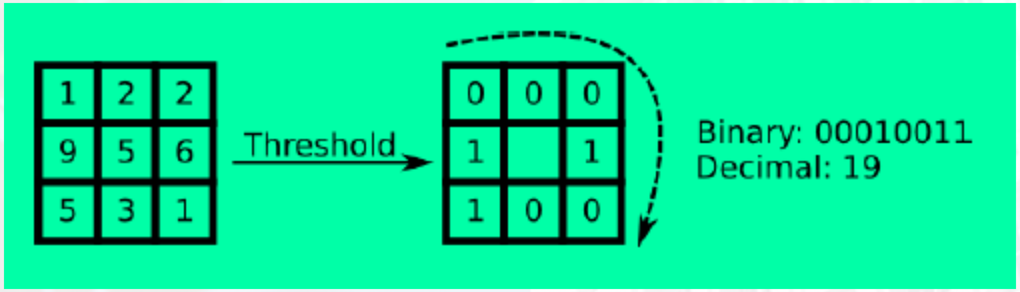
- **Introduction of SIFT**
- **Scale Space**
- **Scale space peak selection**
- **Key point localization**
- **Orientation Assignment**
- **Key point descriptor**
- **Key point matching**
- **HOG**

- LBP

LBP

The LBP operator is defined as a window within the 3*3 window. If the surrounding pixel value is greater than the center pixel value, the position of the pixel is marked as 1, otherwise 0.





SIFT VS HOG VS LBP

Feature	Advantages	Disadvantages	Scope of application
SIFT	1. Maintains invariance for rotation, scale changes, and brightness changes 2. Anti-blocking	Large amount of calculation	Image matching, SFM
HOG	Ignoring the effect of the lighting color on the image, the dimensions of the image that need to characterize the data are reduced.	Descriptive generation slow, low real-time, sensitive to noise	Pedestrian detection
LBP	Not sensitive to light, fast calculation	Sensitive to direction information	Face recognition